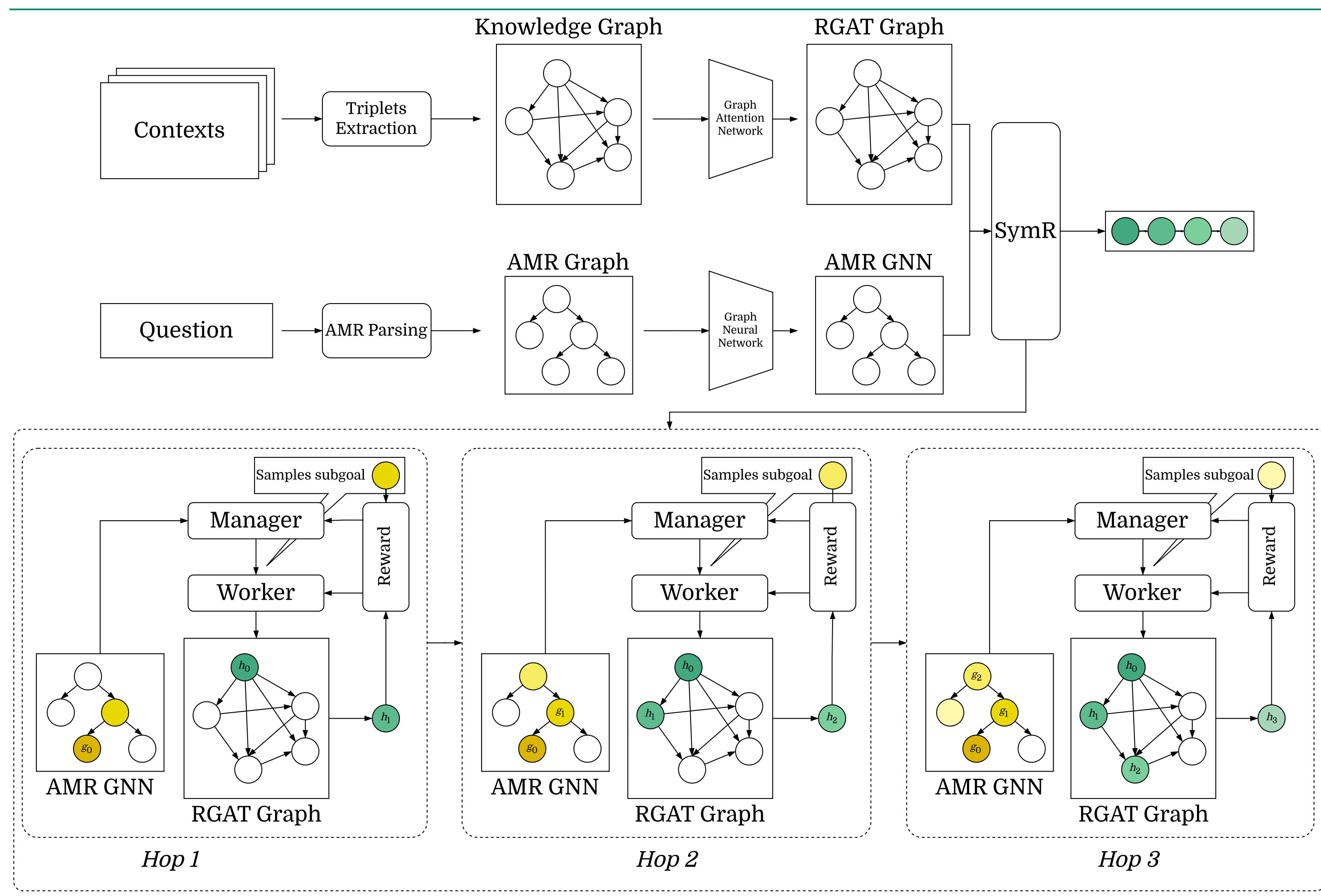


Introduction

Multi-hop Question Answering (MHQA) over Knowledge Graphs requires reasoning across multiple hops, yet existing prompting, decomposition, and search methods often suffer from weak grounding and spurious reasoning; to address this, we propose **SymR**, an AMR-guided hierarchical reinforcement learning framework that generates interpretable symbolic reasoning chains with improved semantic alignment.

SymR Framework Architecture



Question Answering Results

Method	Expl.	Type	EM	F1
SG-Prompt	✓	LLM	61.00	73.30
AnchorCoT	✓	LLM	57.08	66.34
MSRAG	✗	RAG	30.30	30.66
EfficientRAG	✗	RAG	50.59	57.93
HippoRAG	✓	LLM+KG	56.80	69.50
PER-QA	✓	LLM+KG	64.40	62.90
Qwen2.5	✓	LLM	48.46	43.70
Qwen2.5 (KG)	✓	LLM	66.06	53.96
Neural (Ours)	✓	Neural+KG	70.20	74.46
Hierarchy (Ours)	✓	Neural+KG	71.97	75.67
SymR (Ours)	✓	Neural+KG	73.63	75.92

HotpotQA (EM / F1)

Method	Expl.	Type	EM	F1
SG-Prompt	✓	LLM	49.00	57.90
AnchorCoT	✓	LLM	54.42	63.86
MSRAG	✗	RAG	50.80	56.46
EfficientRAG	✗	RAG	44.18	51.64
HippoRAG	✓	LLM+KG	47.30	52.10
PER-QA	✓	LLM+KG	55.40	70.40
Qwen2.5	✓	LLM	52.13	64.11
Qwen2.5 (KG)	✓	LLM	68.00	73.12
Neural (Ours)	✓	Neural+KG	67.77	70.72
Hierarchy (Ours)	✓	Neural+KG	68.77	71.93
SymR (Ours)	✓	Neural+KG	80.91	82.43

2WikiMultihopQA

Method	Expl.	Type	EM	F1
SToC-ToT	✓	LLM	42.00	55.30
AnchorCoT	✓	LLM	41.83	53.31
SeQGraph	✗	RAG	46.01	56.88
HPE	✗	RAG	45.50	53.70
GeAR	✓	LLM+KG	19.00	35.60
HOLMES	✓	LLM+KG	29.00	38.00
Qwen2.5	✓	LLM	22.59	32.46
Qwen2.5 (KG)	✓	LLM	61.68	66.41
Neural (Ours)	✓	Neural+KG	58.42	62.46
Hierarchy (Ours)	✓	Neural+KG	72.51	74.24
SymR (Ours)	✓	Neural+KG	85.83	87.25

MuSiQue (EM / F1)

Core Mathematical Framework

Relational Graph Attention Network

$$\mathbf{h}_i^{(\ell+1)} = \sigma \left(\sum_{j \in \mathcal{N}(i)} \alpha_{ij}^{(\ell)} \mathbf{w}_v^{(\ell)} \mathbf{h}_j^{(\ell)} \right)$$

AMR Encoding & Manager State

$$\mathbf{H}^{(1)} = \text{ReLU}(\tilde{\mathbf{A}}\mathbf{X}^{(0)}\mathbf{W}_1), \quad \mathbf{H}^{(2)} = \tilde{\mathbf{A}}\mathbf{H}^{(1)}\mathbf{W}_2, \quad \mathbf{s}_m^M = [\mathbf{h}_t; \mathbf{p}_t; \mathbf{e}_{\text{target}}]$$

AMR-Guided Subgoal Selection

$$g_0 = \arg \max_{v_i \in \mathcal{V}} \text{sim}(\mathbf{h}_0, \mathbf{H}^{(2)}[i]), \quad g_m \sim \text{Uniform}(\mathcal{N}_{\mathcal{A}}(g_{m-1})), \quad \mathbf{g}_m = \mathbf{H}^{(2)}[g_m]$$

Worker State & Policy

$$\mathbf{s}_t^W = [\mathbf{p}_t; \mathbf{h}_t; \mathbf{g}_{[t/K]}], \quad \mathbf{c}_t = \mathbf{W}_{\text{proj}} \mathbf{s}_t^W, \quad \phi(\mathbf{s}_t^W, \mathbf{h}_j) = \mathbf{w}_W^T \sigma(\mathbf{W}_W[\mathbf{c}_t; \mathbf{h}_j; \mathbf{c}_t \odot \mathbf{h}_j]), \quad \pi_W(a_t = e_j | \mathbf{s}_t^W) = \frac{\exp(\phi(\mathbf{s}_t^W, \mathbf{h}_j))}{\sum_{e_k \in \mathcal{A}_t^{\text{filtered}}} \exp(\phi(\mathbf{s}_t^W, \mathbf{h}_k))}$$

Path Memory & Filtered Action Space

$$\mathbf{p}_t = \text{GRU}(\mathbf{h}_t, \mathbf{p}_{t-1}), \quad \mathcal{A}_t^{\text{filtered}} = \begin{cases} \mathcal{A}_t \setminus \mathcal{V}_t, & |\mathcal{A}_t \setminus \mathcal{V}_t| > 0 \\ \mathcal{A}_t, & \text{otherwise} \end{cases}$$

Hierarchical Reward

$$r_t^{\text{ext}} = \begin{cases} 4.0, & e_T = e^* \\ -0.5, & e_T \neq e^* \\ -0.01, & \text{otherwise} \end{cases} \quad r_t^{\text{sim}} = 0.5 (\text{sim}(\mathbf{h}_{t+1}, \mathbf{e}^*) - \text{sim}(\mathbf{h}_t, \mathbf{e}^*)) \quad r_t^{\text{int}} = \alpha_{\text{int}} (d(\mathbf{h}_t, \mathbf{g}_m) - d(\mathbf{h}_{t+1}, \mathbf{g}_m))$$

PPO Optimisation

$$\delta_t = r_t + \gamma V(s_{t+1}) - V(s_t) \quad A_t = \sum_{l=0}^{T-t-1} (\gamma \lambda)^l \delta_{t+l} \quad \mathcal{L}_W^{\text{CLIP}} = \mathbb{E}_t [\min(\rho_t A_t^W, \text{clip}(\rho_t, 1-\epsilon, 1+\epsilon) A_t^W)] \quad \mathcal{L}_W = -\mathcal{L}_W^{\text{CLIP}} + 0.5 \mathcal{L}_W^V - 0.01 \mathcal{H}[\pi_W]$$

RL Navigation Accuracy

Method	Type	HotpotQA	2Wiki	MuSiQue
Policy Gradient	Neural	46.20	56.83	61.18
Actor-Critic	Value	41.61	54.32	60.59
LSTM-AC	Memory	45.79	64.65	74.10
PPO	Value	47.60	45.91	66.57
PPO-GAE	Deep	46.73	63.31	66.13
DQN	Deep	60.71	79.30	76.09
SymR	Symbolic	75.79	83.29	87.45

Conclusions & Future Work

- **SymR** achieves state-of-the-art performance on HotpotQA, 2WikiMultihopQA, and MuSiQue - outperforming LLM, RAG, and LLM+KG baselines.
- AMR-grounded subgoals *transform sparse rewards into dense intrinsic signals*, accelerating convergence and reducing spurious paths.
- The *two-stage* design (SymR path → LLM verbalization) isolates reasoning quality from language generation, ensuring interpretability.
- **Future:** (1) End-to-end learned AMR parsing; (2) Dynamic/incomplete KG settings; (3) Richer symbolic constraints (logic, text evidence).

Acknowledgement

This research is funded under grant number SVOISP-2025-KH & KTMT-06. We acknowledge Ho Chi Minh City University of Technology (HCMUT), VNU-HCM, for supporting this study.